

Practical -02

a) Code output:

```
lenovo@DESKTOP-NTSK193:~$ nano pract2.c
lenovo@DESKTOP-NTSK193:~$ gcc pract2.c
lenovo@DESKTOP-NTSK193:~$ ./a.out
Files successfully opened
File copied successfully
lenovo@DESKTOP-NTSK193:~$
```

Contents of source file

```
lenovo@DESKTOP-NTSK193:~$ cat source.txt
This is the Practical 2 Session
University KLH
lenovo@DESKTOP-NTSK193:~$
```

Contents of destination file

```
lenovo@DESKTOP-NTSK193:~$ cat destination.txt
This is the Practical 2 Session
University KLH
lenovo@DESKTOP-NTSK193:~$
```

b) Command : strace

```
lenovo@DESKTOP-NTSK193: ~ % strace cat /etc/passwd
```

```
execve("/usr/bin/cat", ["cat", "passwd"], 0x7ffcf22d54c8 /* 25 vars */) = 0  
brk(NULL)                               = 0x58672f49e000  
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x769ffa624000  
access("/etc/lid.so.preload", R_OK)     = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/etc/lid.so.cache", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=27891, ...}) = 0  
mmap(NULL, 27891, PROT_READ, MAP_PRIVATE, 3, 0) = 0x769ffa61d000  
close(3)                                = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libss.so.1", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0  
mmap(NULL, 206208, PROT_READ, MAP_PRIVATE, MAP_DENYWRITE, 3, 0) = 0x769ffa5ea000  
mmap(0x769ffa5f1000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x769ffa5f1000  
mmap(0x769ffa612000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x769ffa612000  
mmap(0x769ffa619000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x769ffa619000  
mmap(0x769ffa61b000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x769ffa61b000  
close(3)                                = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0  
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE, MAP_DENYWRITE, 3, 0) = 0x769ffa5bc000  
mmap(0x769ffa5c0000, 174756, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x769ffa5c0000  
mmap(0x769ffa5e4000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x769ffa5e4000  
mmap(0x769ffa5e8000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x769ffa5e8000  
close(3)                                = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libnss.so.6", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0  
mmap(NULL, 1208152, PROT_READ, MAP_PRIVATE, MAP_DENYWRITE, 3, 0) = 0x769ffa496000  
mmap(0x769ffa4a8000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x769ffa4a8000  
mmap(0x769ffa548000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x769ffa548000  
mmap(0x769ffa5ba000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x769ffa5ba000  
close(3)                                = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
```

[illegible]

